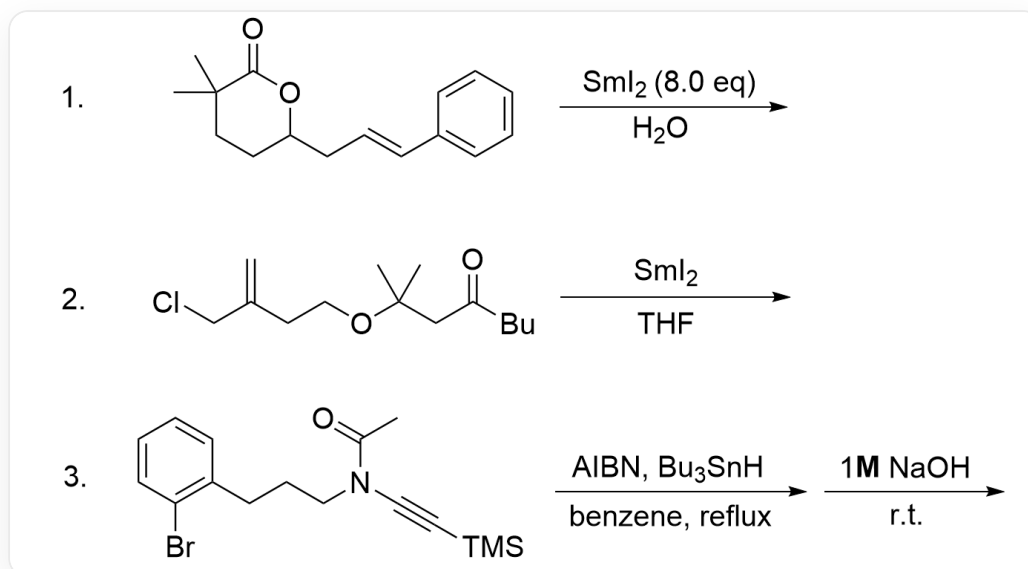


Question

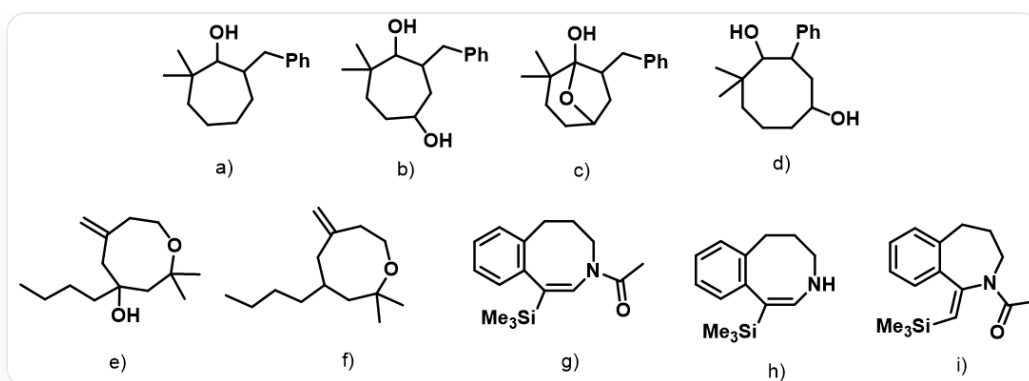
The construction of medium-sized rings is different from the construction of five- or six-membered rings. Due to their larger ring strain, some relatively more special reaction conditions are required to achieve this. Radical chemistry can play a good role in the construction of this type of ring system. By using simple reduction conditions, medium-sized rings can also be constructed efficiently. Select the final products of the following 3 reactions, without requiring stereochemistry.



Reaction 1: CC1(C)CCC(C/C=C/C2=CC=CC=C2)OC1=O reacts with SmI_2 in H_2O

Reaction 2: C=C(CCOC(C)(C)CC(CCCC)=O)CCl reacts with SmI_2 in THF

Reaction 3: BrC1=CC=CC=C1CCCN(C(C)=O)C#C[Si](C)(C)C reacts with Bu_3SnH , $AIBN$, $NaOH$ in benzene under reflux



a) CC(C1O)(C)CCCCC1CC2=CC=CC=C2 b) OC(CC1CC2=CC=CC=C2)CCC(C)(C)C1O c) CC1(C)CCC(CC2CC3=CC=CC=C3)OC12O d) CC1(C)CCCC(O)CC(C2=CC=CC=C2)C1O e) C=C1CCOC(C)(C)CC(CCCC)(O)C1 f) C=C1CCOC(C)(C)CC(CCCC)C1 g) CC(N1/C=C([Si](C)(C)C)C2=CC=CC=C2)CCC1=O h) C[Si](C)(C)C1=CC=CC=C1NCCC2=CC=CC=C2 i) O=C(C)N1/C(C2=CC=CC=C2CCC1)=C/[Si](C)(C)C

A. All other options are incorrect

B. a.e.g.

C. c.e.g.

D. b.e.g.

E. d.e.h.

F. b.f.g.

G. c.e.i.

H. d.e.i.

I. b.e.h.

Answer

Correct Answer: **D**

Detailed Explanation

Reaction 1:

CC1(C)CCC(C/C=C/C2=CC=CC=C2)OC1=O reacts with SmI_2 in H_2O , the carbonyl group is reduced to a hydroxyl group and a carbon radical, undergoing a 5-exo ring closure, to obtain OC(CC1C[X, X is a single electron]C2=CC=CC=C2)CCC(C)(C)C1O

CHECKPOINT

1 PTS

OC(CC1C[X, X is a single electron]C2=CC=CC=C2)CCC(C)(C)C1O

OC(CC1C[X, X is a single electron]C2=CC=CC=C2)CCC(C)(C)C1O is further reduced and quenched with water to obtain c.

CHECKPOINT

1 PTS

OC(CC1C[X, X is a single electron]C2=CC=CC=C2)CCC(C)(C)C1O is further reduced and quenched with water to obtain c.

c can undergo ring-opening in water and is further reduced to obtain b.

CHECKPOINT

1 PTS

c can undergo ring-opening in water and is further reduced to obtain b.

Reaction 2:

Allyl chloride is reduced to an allyl radical, and the carbonyl group is also reduced to a carbon radical and an oxygen anion. The two radicals couple and quench to obtain e.

CHECKPOINT

1 PTS

Reaction 2, allyl chloride is reduced to an allyl radical, and the carbonyl group is also reduced to a carbon radical and an oxygen anion. The two radicals couple and quench to obtain e.

Reaction 3:

Under radical initiation conditions, Sn radicals are generated, abstracting Br to obtain phenyl radicals, undergoing 8-endo ring closure on the alkyne to obtain CC(N1/C[X, X is a single electron]=C([Si](C)(C)C)\C2=C(C=CC=C2)CCC1)=O

CC(N1/C[X, X is a single electron]=C([Si](C)(C)C)\C2=C(C=CC=C2)CCC1)=O is quenched by $Sn-H$ to obtain g.

Because Si can stabilize β -position radicals, 7-endo ring closure does not occur.

The amide bond is difficult to hydrolyze at room temperature, and the acetyl group will not leave, so the product is not h.

CHECKPOINT

1 PTS

Because Si can stabilize β -position radicals, 7-endo ring closure does not occur.

CHECKPOINT

1 PTS

Undergoing 8-endo ring closure to obtain CC(N1/C[X, X is a single electron]=C([Si](C)(C)C)\C2=C(C=CC=C2)CCC1)=O

CHECKPOINT

1 PTS

CC(N1/C[X, X is a single electron]=C([Si](C)(C)C)\C2=C(C=CC=C2)CCC1)=O is quenched by Sn-H to obtain g.

CHECKPOINT

1 PTS

The amide bond is difficult to hydrolyze at room temperature, the product is not h

In summary, option D is reasonable.